

FSI RE:RASSOR BLDC Gearbox Assembly

Instructions and Datasheet

Specs:

Weight (no motor)	~ 27 grams
Gear ratio	3.769 : 1
Tooth count:(ring \ planet \ sun)	49 / 18 / 13
Intended input RPM range:	0 - 1000
Lubrication Type:	Silicone Grease (optional)
Intended Filament:	PETG

Additional Hardware list:

M2 screws 5mm (+/- 1) length	4x (cap)
M2 screws 8mm length	3x (sun gear)
M4 screws 4mm length	8x (motor mount and encoder board)
M4 screws 8mm length	8x (for cycloidal gear mounting)

Feature overview:

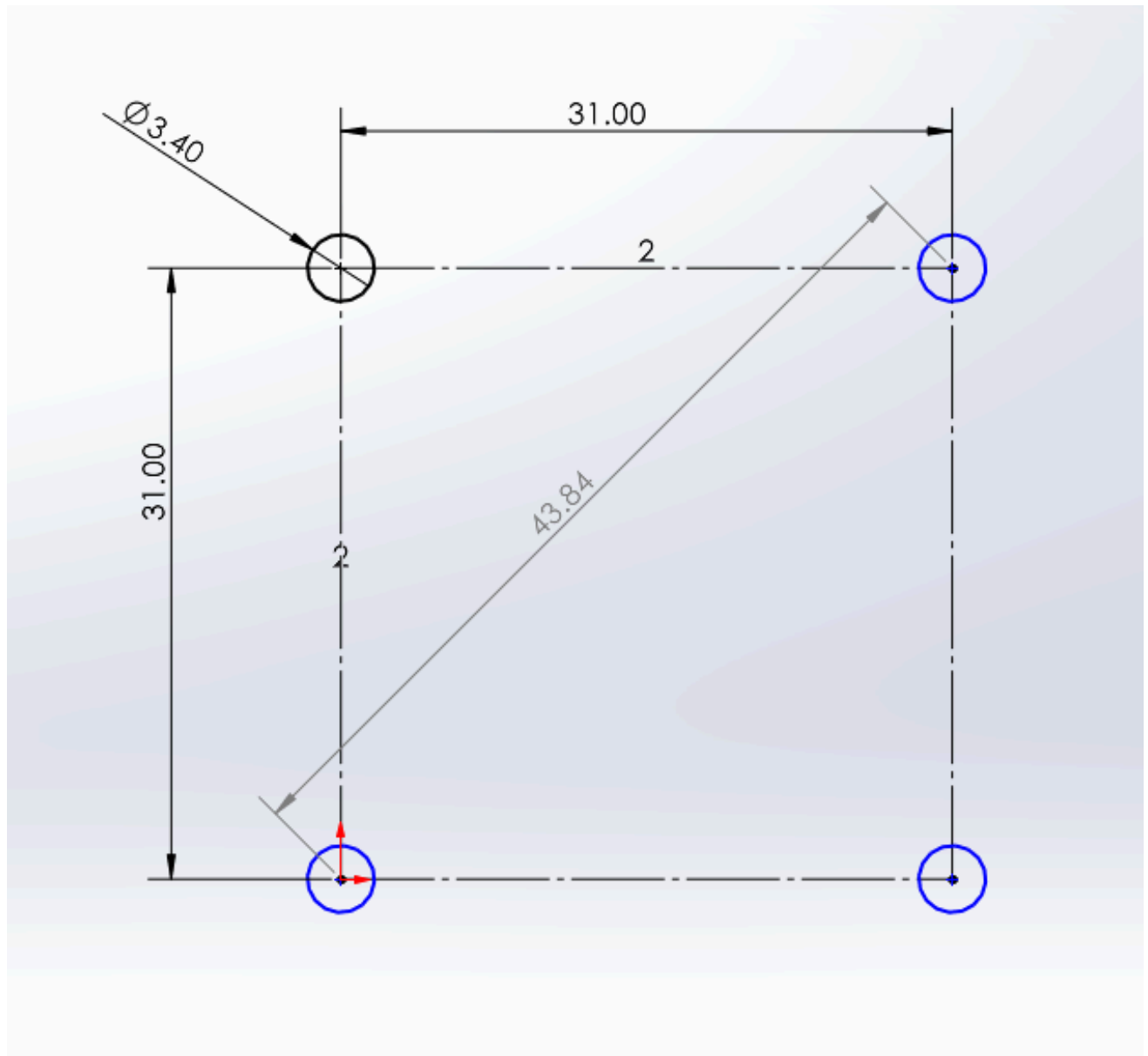
This gearbox was designed specifically for 3d printing. No additional parts are needed besides screws to assemble. All gear pairs tooth counts are co-prime, and the tooth pressure angle was increased to 30 degrees. These Features were chosen to increase wear life and efficiency. This gearbox is intended for use with the FSI RE:RASSOR BLDC motor system (SD2 Spring 2026).

Printing Notes:

It is recommended to print the “BY_OBJECT” plates. From my experience, they have much better tolerances and surface finish creating a much smoother operation.

This system was prototyped and printed on a Bambu Labs P1S. My printer has a pretty consistent +0.005in on the final part compared to the CAD models. This may vary from printer to printer. Since the bearing surfaces are 3d printed, in order to reduce play in the system, a very very slight interference fit is most desirable on the “output shaft bushing”, although not mandatory. The print includes a variety of drive shaft bushings with slightly over or under sized bushings (printed on plate 2), find the best fit for your gearbox on your printer by testing them. If in doubt use the default size (prints on plate 1).

Mounting hole configuration (in mm):

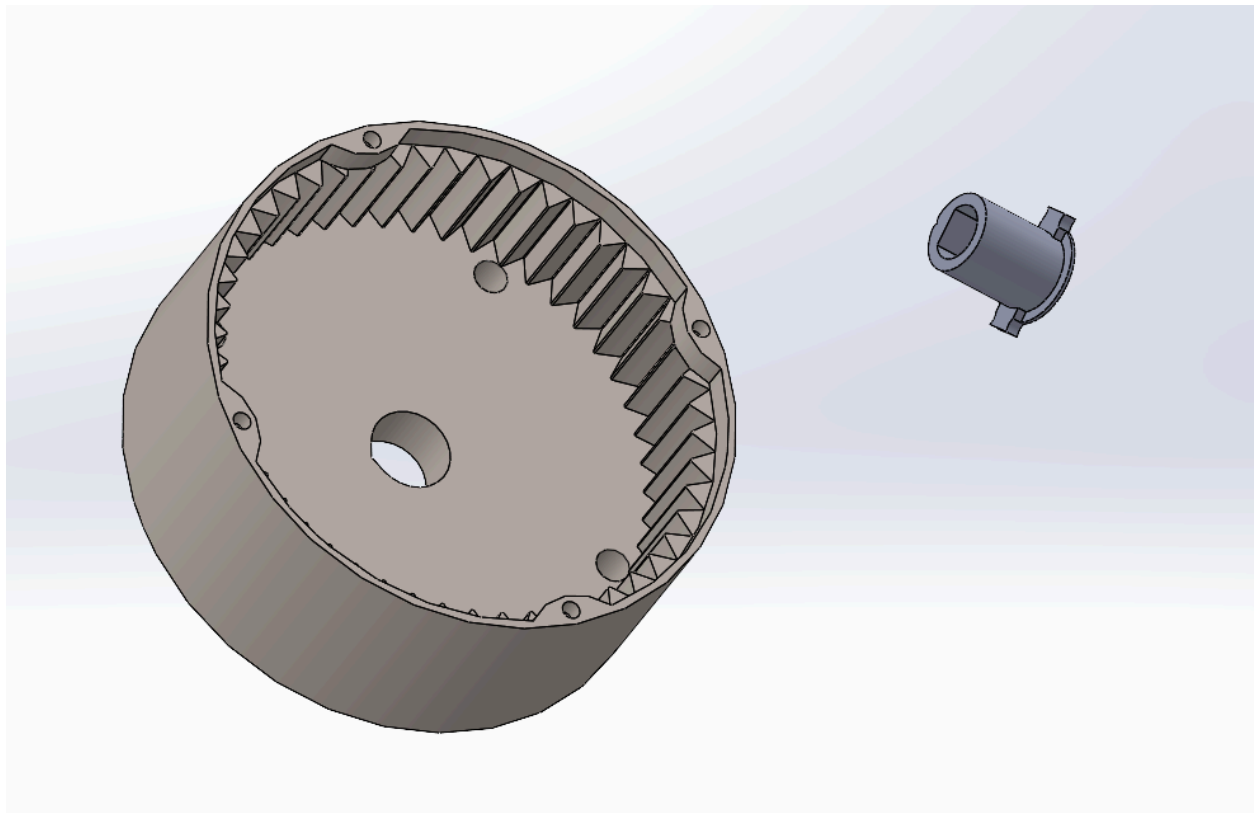


Assembly Instructions

Step 1:

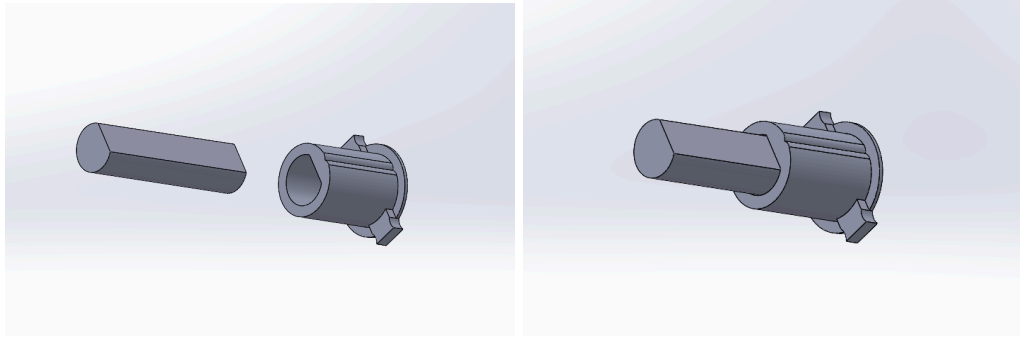
Print all of the parts in the gear box .3mf file. Test the fit of the “drive shaft bushing” with both the “Ring Gear” and “Planet Carrier” parts. It should be free spinning but with as little free play as possible.

There are multiple sizes of driveshaft bushing provided on plate 2

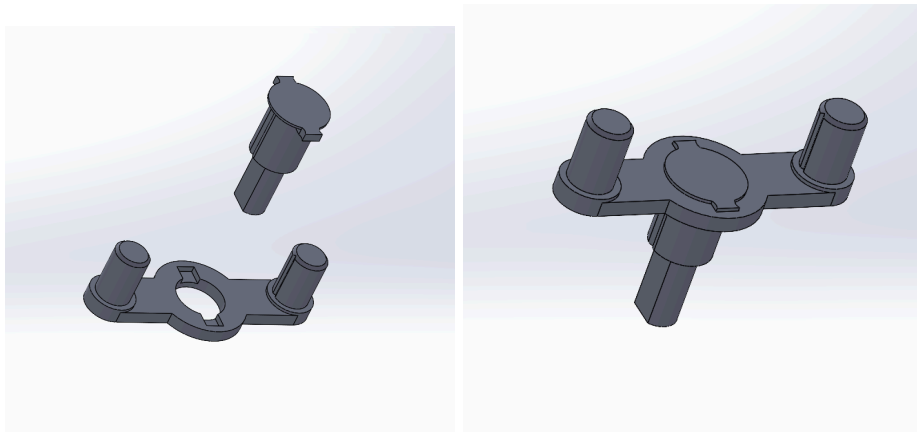


Step2:

Assemble the drive shaft and the drive shaft bushing. Secure with SuperGlue

**Step 3:**

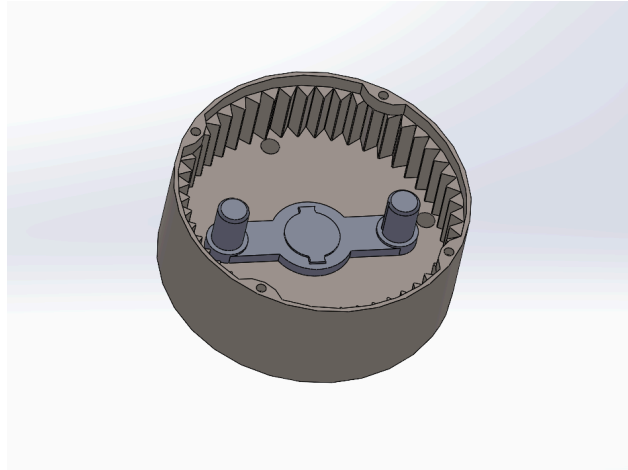
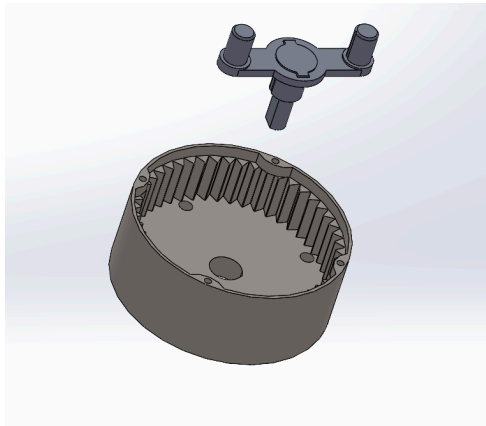
Assemble the drive shaft assembly with the planet carrier. Note that the bushing has a key. Use superglue to secure.



Step 4:

Apply silicone grease to the cylindrical surface of the drive shaft bushing and then assemble with the ring gear.

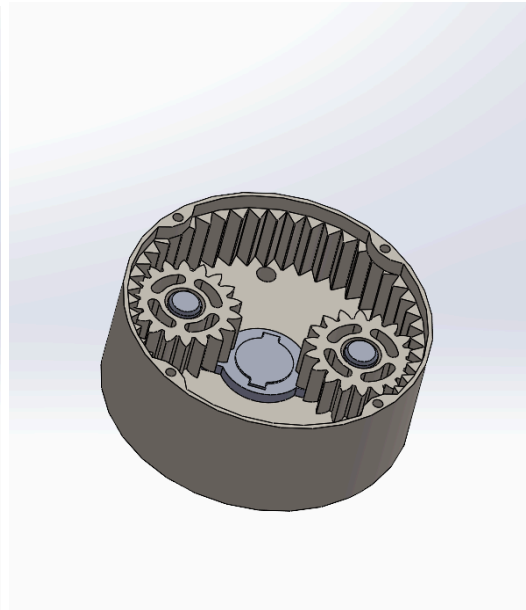
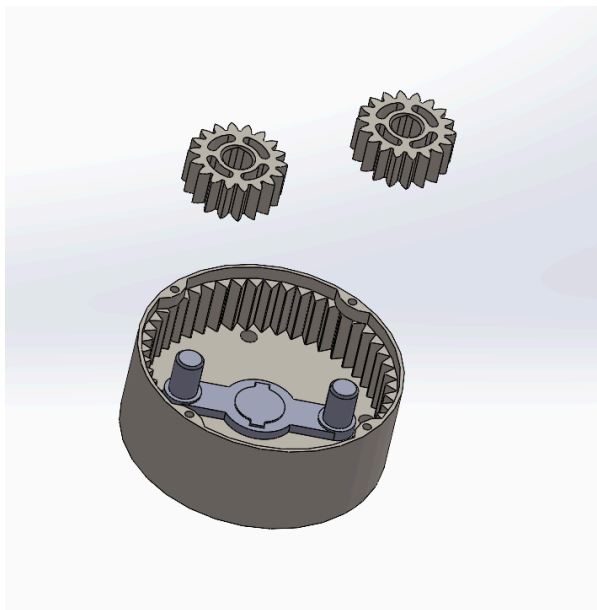
The fit should be free spinning, but as close of a fit as possible. This is the main bearing surface. It should be able to be pushed in by hand, and spun by hand after assembled



Step 5:

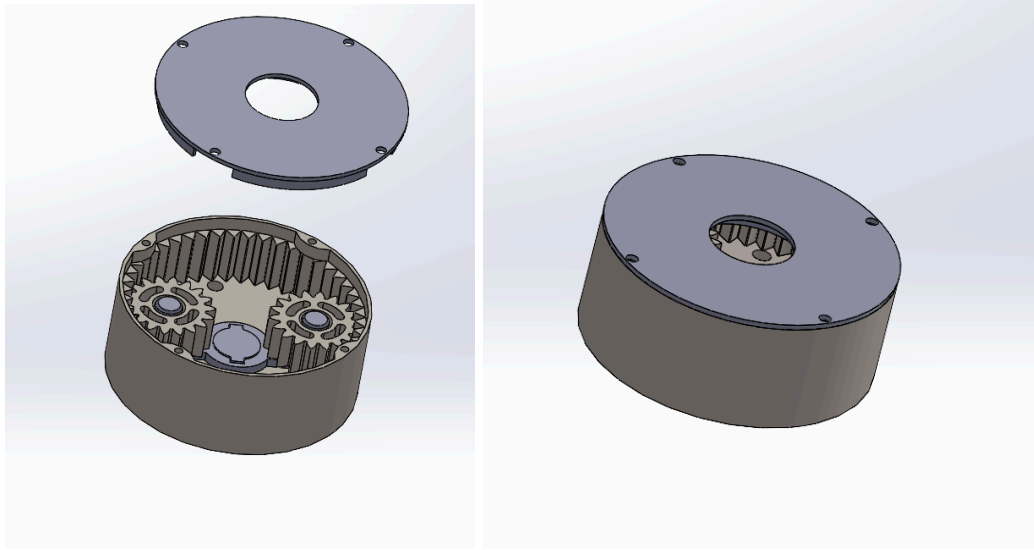
Add the planet gears onto the pins of the planet carrier. You may have to rotate the planet carrier out of the way of the cap mounting holes in order to do this.

Apply grease to the teeth of the ring gear once assembled



Step 6:

Add the cap and screw it in with the M2 x 4mm screws



[to be continued for the motor mount]

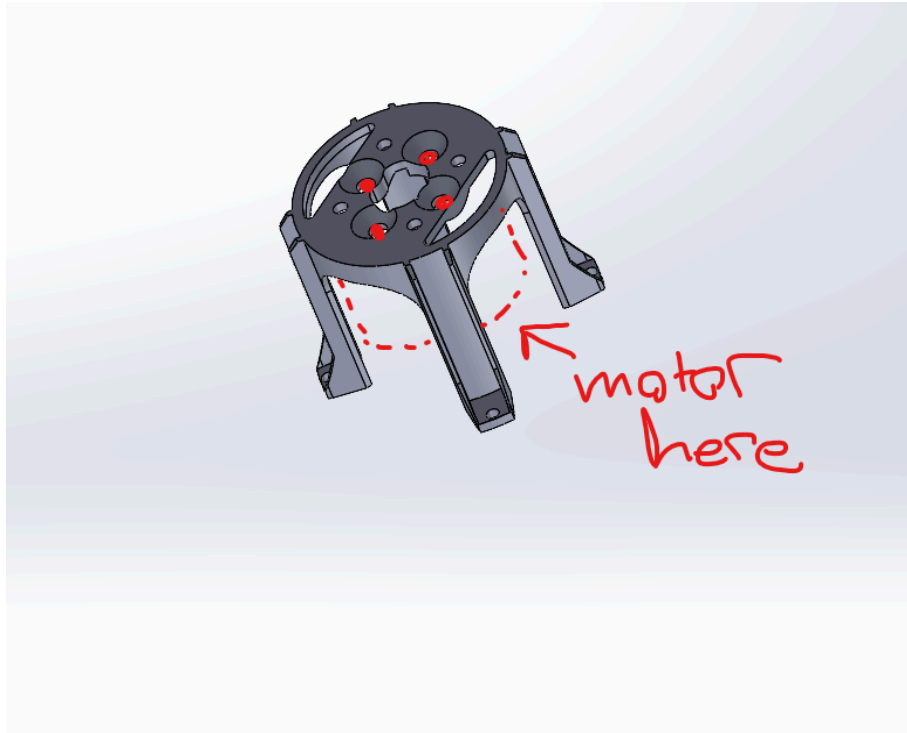
Step 7:

Screw the sun gear to the motor



Step 8:

Screw the motor to the motor mount using the m4 screws



Step 9:

Screw the motor mount to the Ring Gear through the holes in the cap

